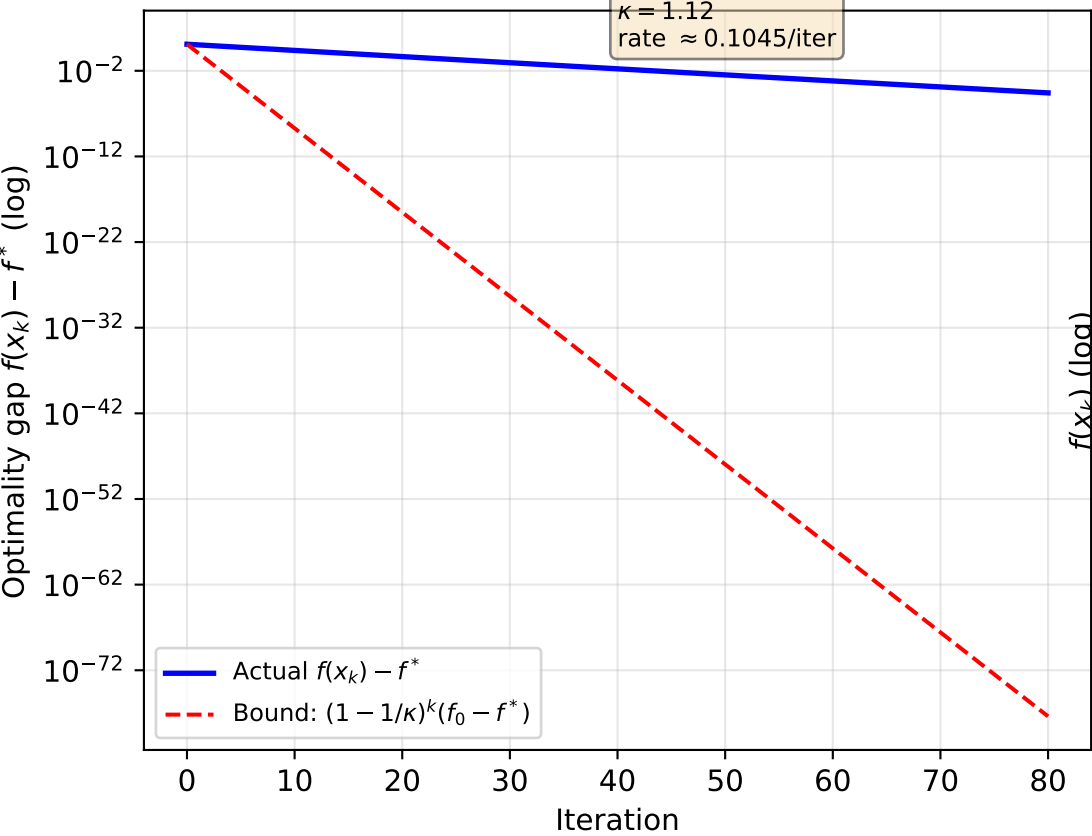
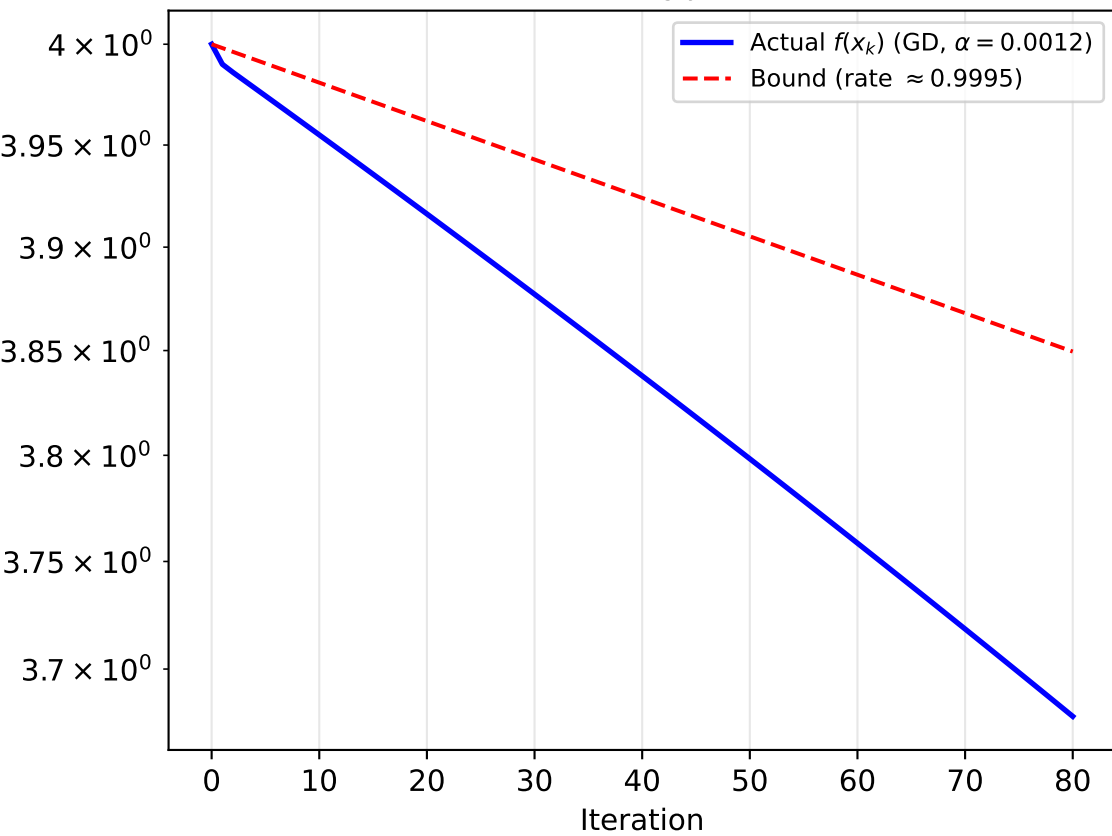


GD Convergence: Theory vs Practice

Benchmark A: Actual vs Theoretical Bound
($\kappa = 1.12, \alpha = 0.08$)



Benchmark C: GD Stagnation on Rosenbrock
($\alpha = 0.0012, L_{\text{start}} \approx 1002$)



Near-flat:
rate ≈ 0.9995
per iteration